Lecture 12 (10/12)

Study design, confounding, and multiple predictors

Unit 3 summary:

- An estimator $\hat{\theta}$ is how we approximate a parameter θ from data. Its variation over replicates of the data (repeated experiments, different samples) is the sample distribution.
- The error breaks down into bias and variance, $\text{MSE}_{\hat{\theta}} = \text{se}(\hat{\theta})^2 + \text{Bias}^2$. We look for consistent estimators where both vanish.
- Confidence intervals and p-values are computed from the sample distribution.
- The least squares estimators $\hat{\beta}_0, \hat{\beta}_1$ provide unbiased estimates of parameters in a regression model (assuming data points are sampled independently). $R^2 = \hat{\rho}^2$ is the estimate of ρ^2.
- Autoregressive models (specifically AR(1) models) are a type of regression model where the next value in the time series is the response variable. Here the regression coefficient is the correlation and as ρ approaches 1 the long-term variance diverges
- In AR(1) models, the (X, Y) points are successive values (Y_t, Y_{t+1}) and the pairs are therefore not independent, leading to a situation where least squares estimators are biased (you don't have to know the formula, but understand why it is biased).

Unit 4: regression with multiple predictors.

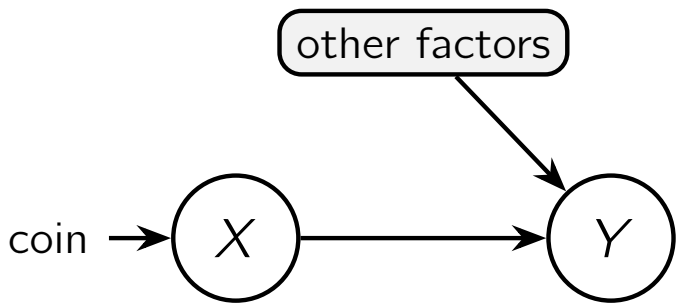
- 4.1 Study design and confounding (today – briefly).
- 4.2 Multiple-predictor regression and its interpretation (today).
- 4.3–4.4 Estimating the coefficients, their sample distribution, collinearity.
- 4.5 Categorical predictors.

Two ways to collect data

Does a treatment ($X = 1$) change an outcome Y, compared with no treatment ($X = 0$)?

Randomized controlled trial (RCT)

we assign X by a coin flip



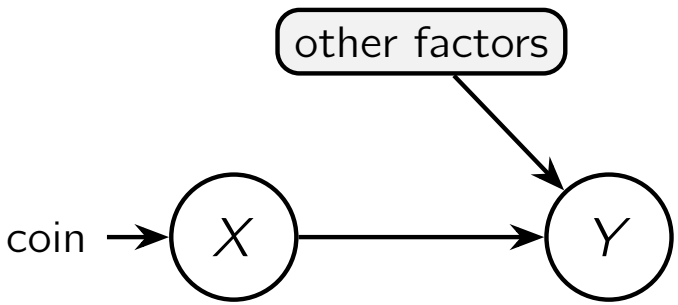
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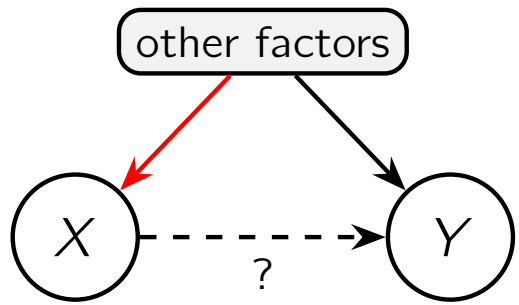
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Cohort study (observational)

we follow people and observe who gets X



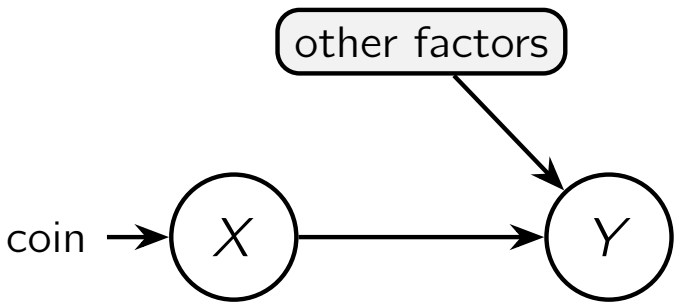
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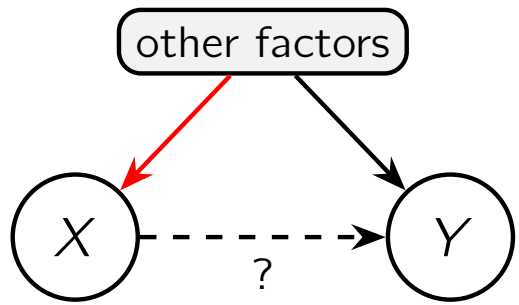
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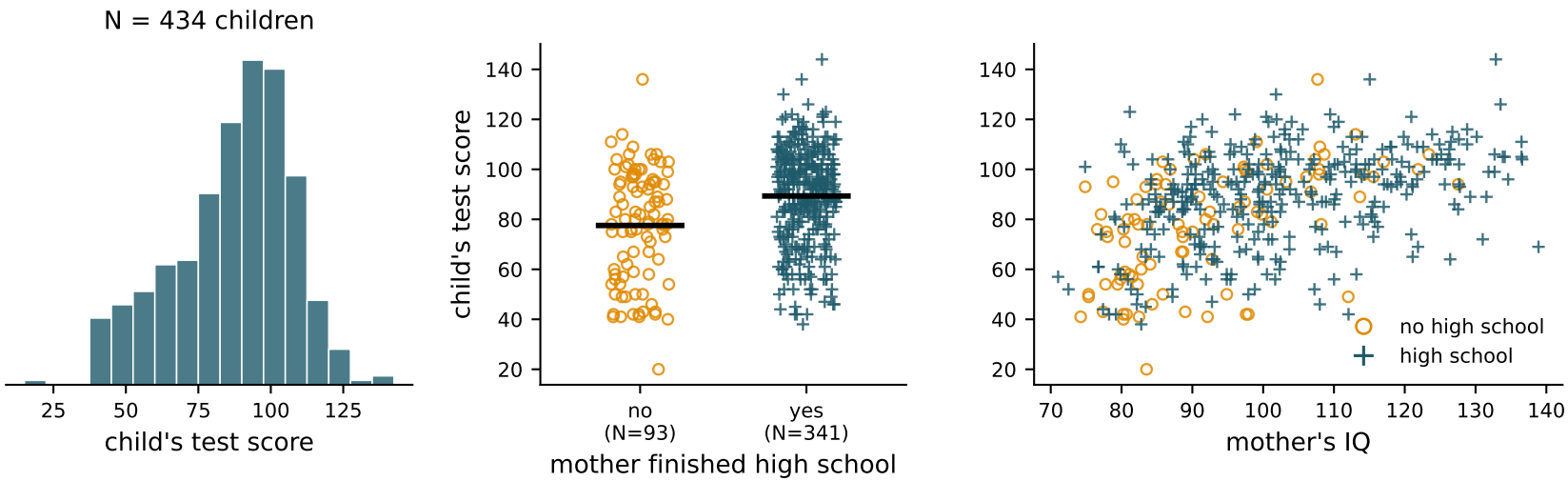
X may be correlated with other factors that also affect Y, so we cannot tell what causes what.

For most questions in the social sciences, an RCT is impossible.

Children's test scores

Test scores of $N = 434$ children, with two facts about each child's mother:

- Y: the child's test score;
- $X_{\text{hs}} = 1$ if the mother finished high school, 0 if not;
- X_{iq}: the mother's IQ.[1]

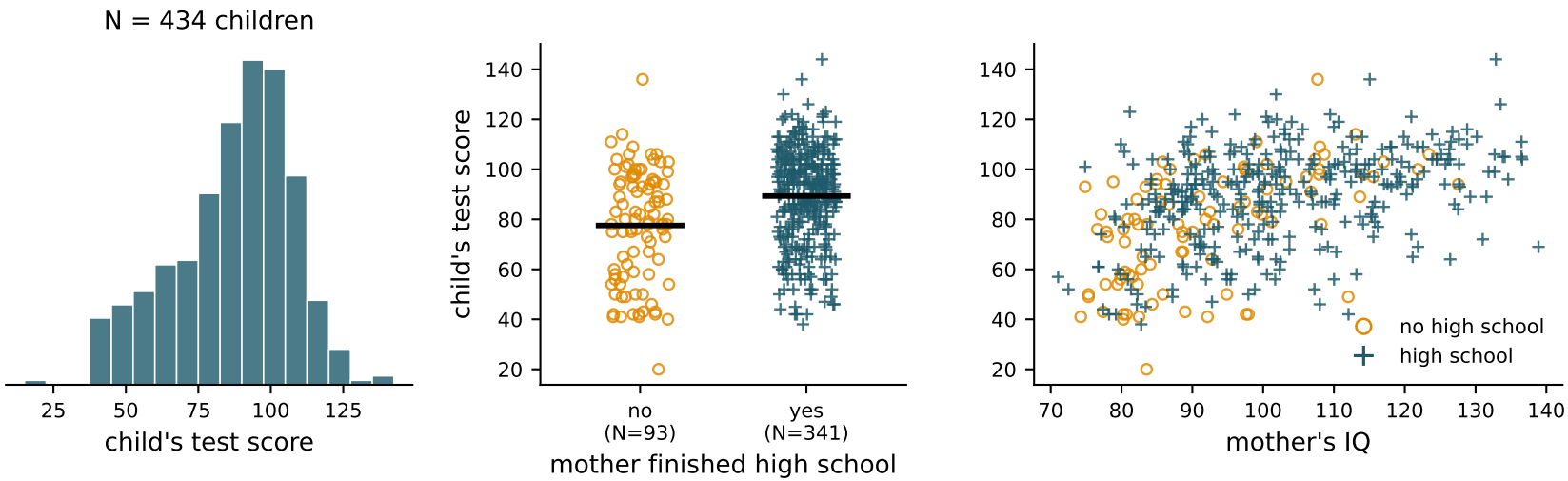


[1] IQ is an imperfect measure of cognitive ability: it is a single test score, and it also depends on the particular test, the testing conditions, and the person's education and background.

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Question

From the right two plots, estimate σ_{IQ}, σ_{HS}, σ_Y as well as the regression coefficients and correlation coefficients.

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Children's test scores

Standard deviations ($N = 434$):

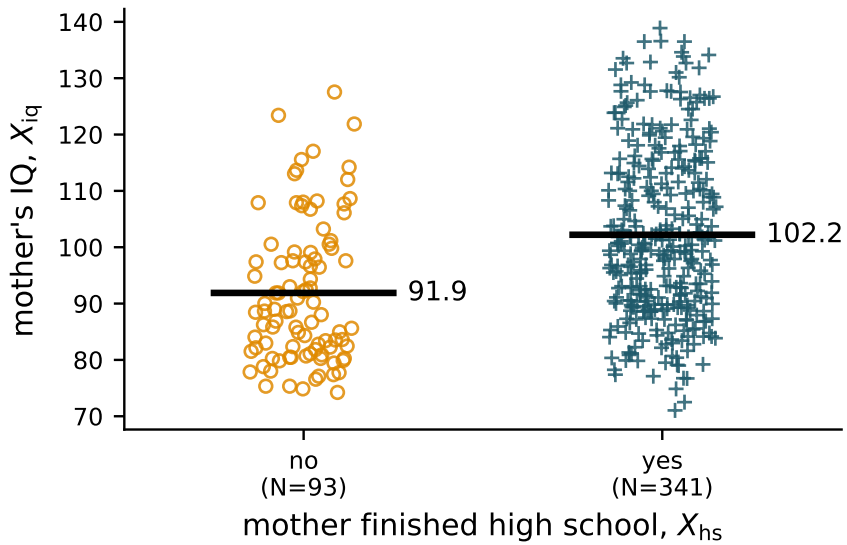
	standard deviation
Y (child's score)	$\sigma_Y = 20.4$
X_{hs}	$\sigma_{\text{HS}} = 0.41$
X_{iq}	$\sigma_{\text{IQ}} = 15$

Single-predictor regressions. We write β' (with a prime) for a single-predictor coefficient, to distinguish it from the coefficients β of the two-predictor model we will fit later:

- Y on X_{hs}: $\hat{\beta}'_{\text{hs}} = 11.8$ (the difference in average score between the two groups), $\hat{\rho} = \hat{\beta}'_{\text{hs}}\sigma_{\text{HS}}/\sigma_Y \approx 0.24$.
- Y on X_{iq}: $\hat{\beta}'_{\text{iq}} = 0.61$, $\hat{\rho} = \hat{\beta}'_{\text{iq}}\sigma_{\text{IQ}}/\sigma_Y \approx 0.45$.

IQ and HS

Mother's IQ vs. whether she finished high school:



What we computed vs. what we want

- **Computed from single predictor regression**: the difference in average score between the groups,

$$\beta'_{\text{hs}} = E[Y \mid X_{\text{hs}} = 1] - E[Y \mid X_{\text{hs}} = 0] \qquad (\hat{\beta}'_{\text{hs}} = 11.8).$$

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$$\beta_{\text{hs}} = E[Y \mid X_{\text{hs}} = 1, X_{\text{iq}} = x] - E[Y \mid X_{\text{hs}} = 0, X_{\text{iq}} = x]$$

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- To compare children whose mothers have the *same* IQ, put both variables in the model:

$$Y = \beta_0 + \beta_{\text{hs}} X_{\text{hs}} + \beta_{\text{iq}} X_{\text{iq}} + \epsilon, \qquad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2).$$

Fitting the model in Python: `statsmodels`

The `statsmodels` package fits regression models and reports the estimates and their standard errors:

```
import statsmodels.api as sm

X = sm.add_constant(np.column_stack([mom_hs, mom_iq]))
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No. Observations:                 434   Adj. R-squared:                  0.210
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         25.7315      5.875      4.380      0.000      14.184      37.279
mom_hs         5.9501      2.212      2.690      0.007       1.603      10.297
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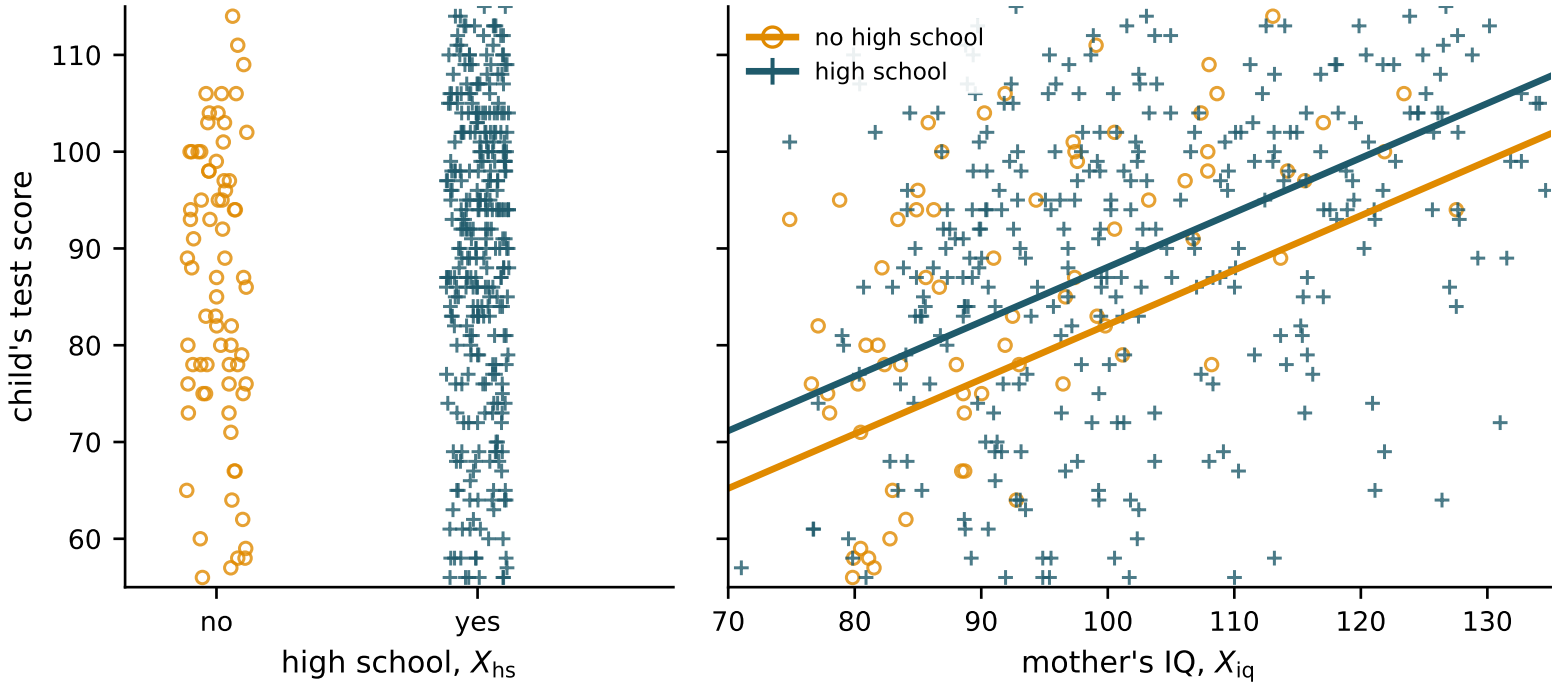
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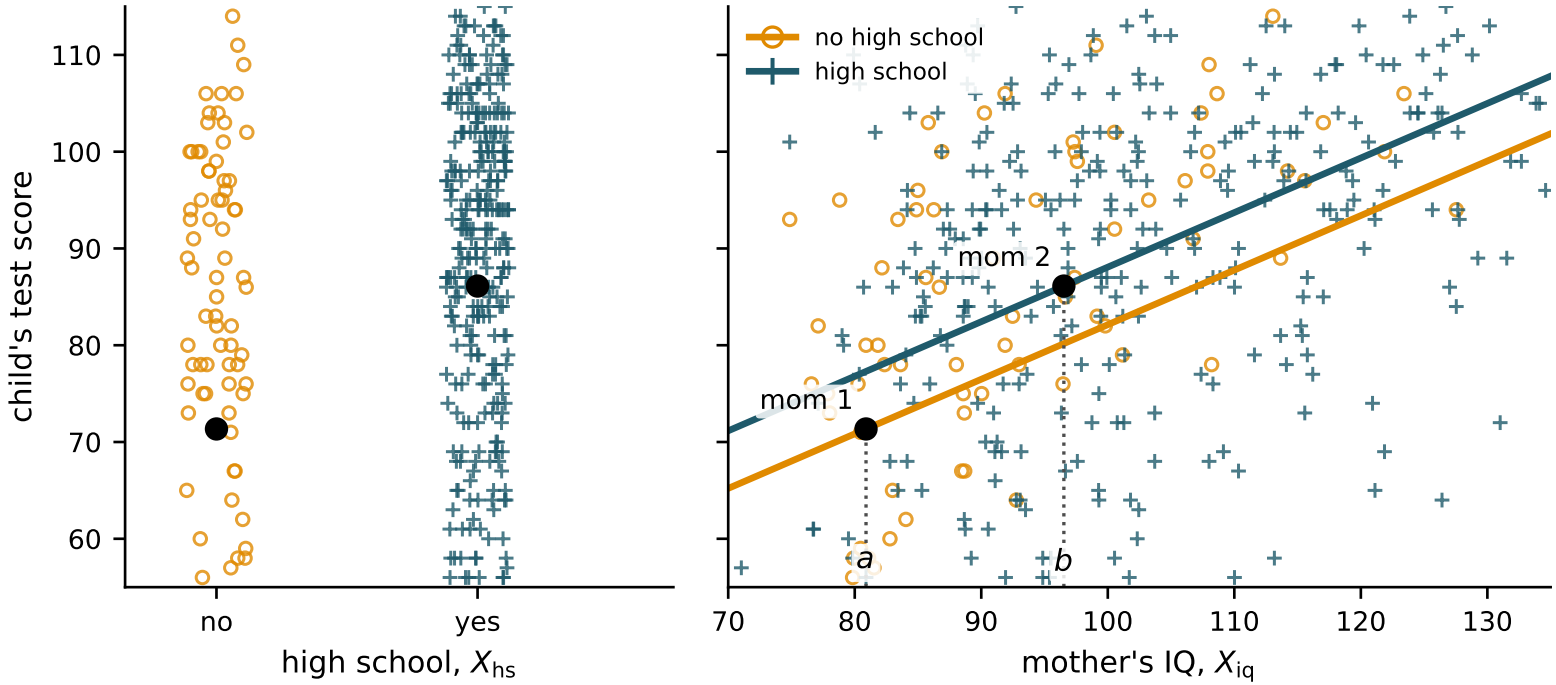
The `coef` column gives the least-squares estimates: $\hat{\beta}_0 = 25.7$, $\hat{\beta}_{\text{hs}} = 5.95$, $\hat{\beta}_{\text{iq}} = 0.56$. (`add_constant` adds the intercept β_0; the other columns come later.)

From two predictors to one



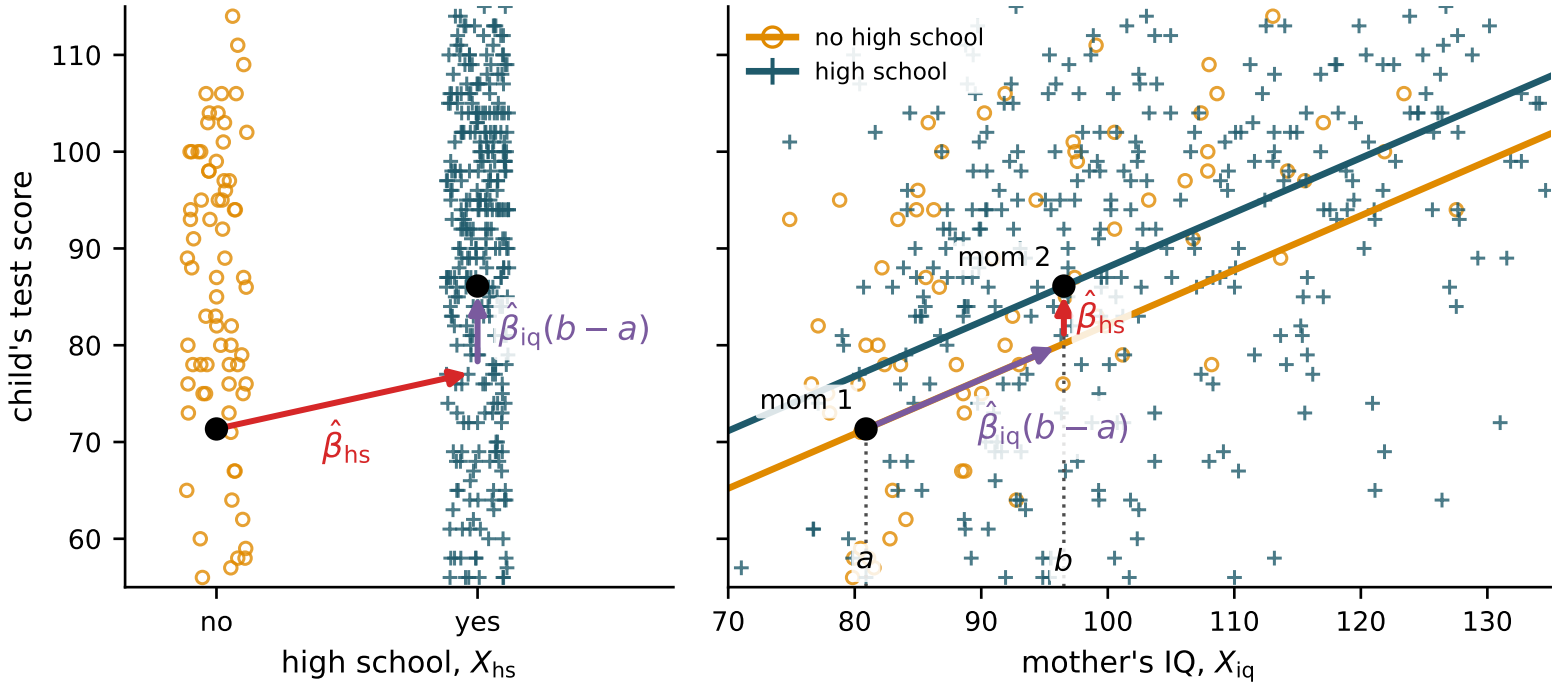
The fitted two-predictor model, vs. schooling (left) and vs. IQ (right).

From two predictors to one



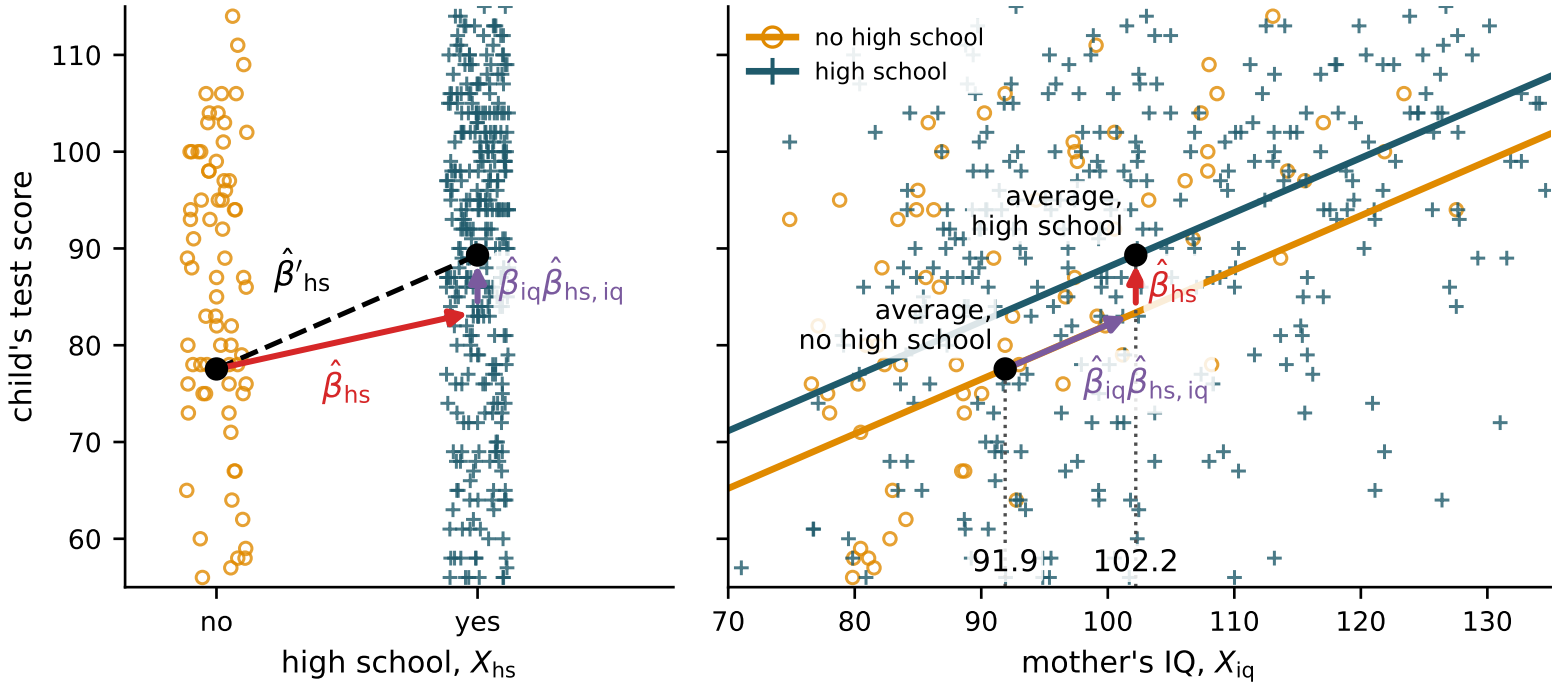
Pick a random mom without high school (IQ $= a$) and one with (IQ $= b$).

From two predictors to one



Their difference in expected score is β_{hs} (schooling, IQ fixed) + $\beta_{\text{iq}}(b-a)$ (difference in IQ).

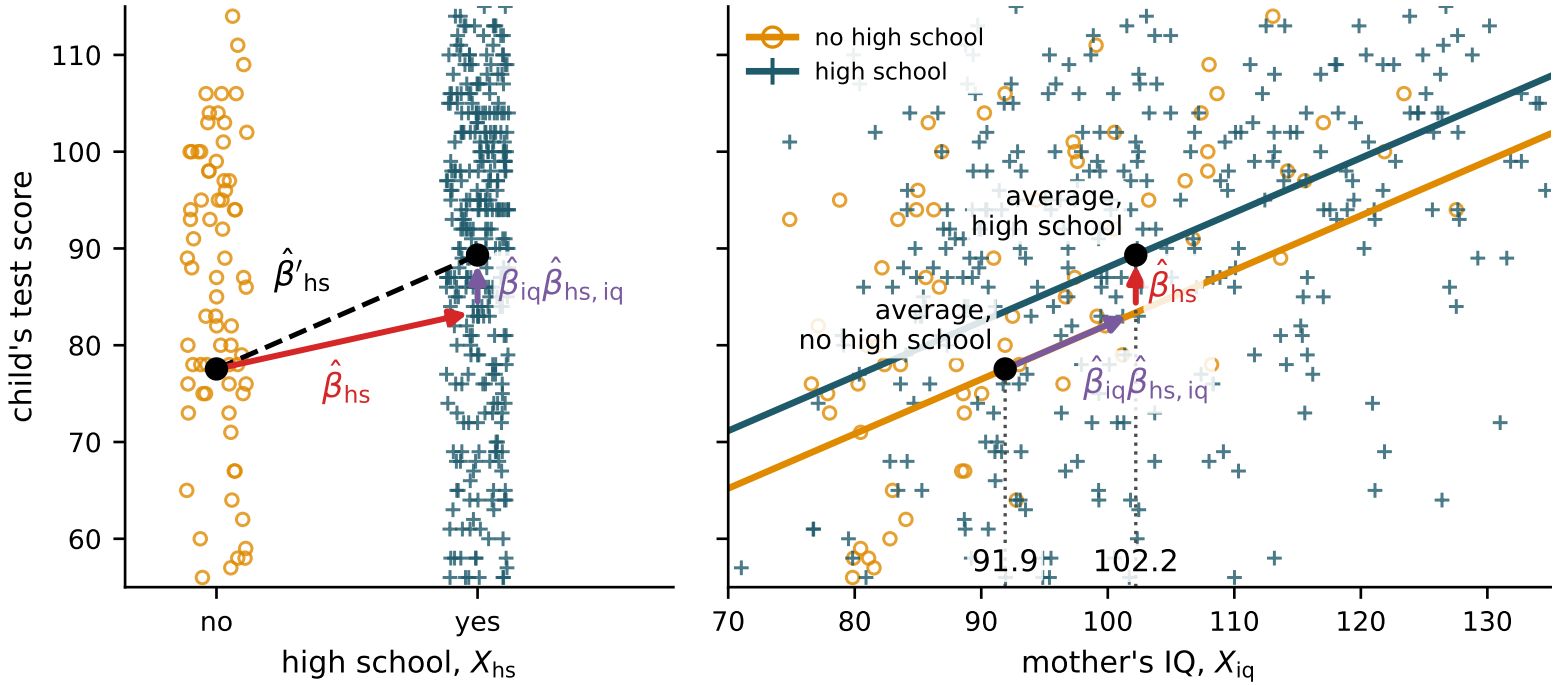
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Average over all moms in each group: a and b are replaced by the average IQs,

$$\underbrace{E[Y \mid X_{\text{hs}} = 1] - E[Y \mid X_{\text{hs}} = 0]}_{\beta'_{\text{hs}}} = \beta_{\text{hs}} + \beta_{\text{iq}} \underbrace{\big(E[X_{\text{iq}} \mid X_{\text{hs}} = 1] - E[X_{\text{iq}} \mid X_{\text{hs}} = 0]\big)}_{\beta_{\text{hs,iq}}}$$

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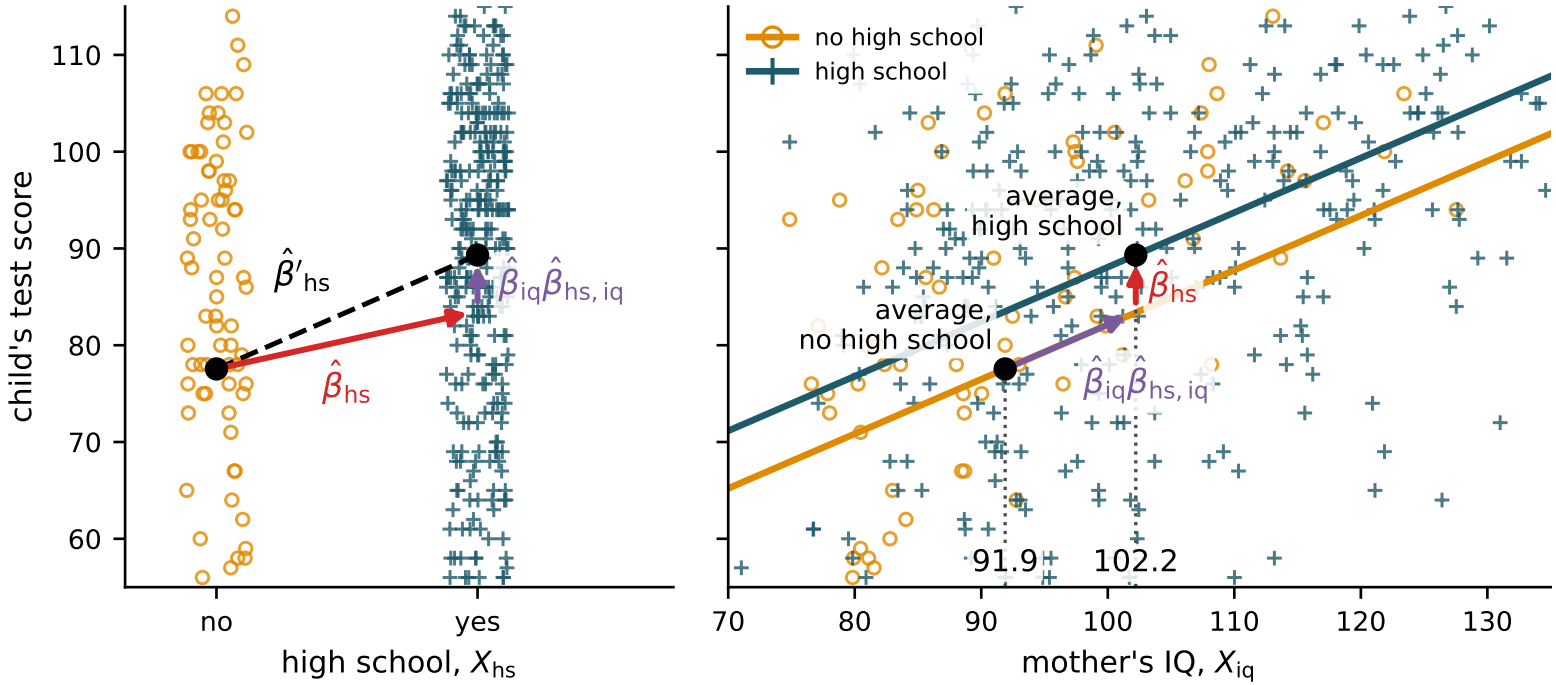


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so $\beta'_{hs} = \beta_{hs} + \beta_{iq}\beta_{hs,iq}$: $11.8 = 5.95 + 0.56 \times 10.3$.

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so $\beta'_{\text{hs}} = \beta_{\text{hs}} + \beta_{\text{iq}}\beta_{\text{hs,iq}}$: $11.8 = 5.95 + 0.56 \times 10.3$.
We say β_{hs} is the effect of schooling controlling for IQ (and vice versa).

Which predictor matters more?

$\hat{\beta}_{\text{hs}} = 5.95$ is ten times $\hat{\beta}_{\text{iq}} = 0.56$, but the coefficients have different units (points for finishing high school vs. points per IQ point), so we cannot compare them directly.

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Instead, compare the difference in average score for a *typical* change in each predictor, one standard deviation: $\hat{\beta}_k \sigma_{X_k}$.

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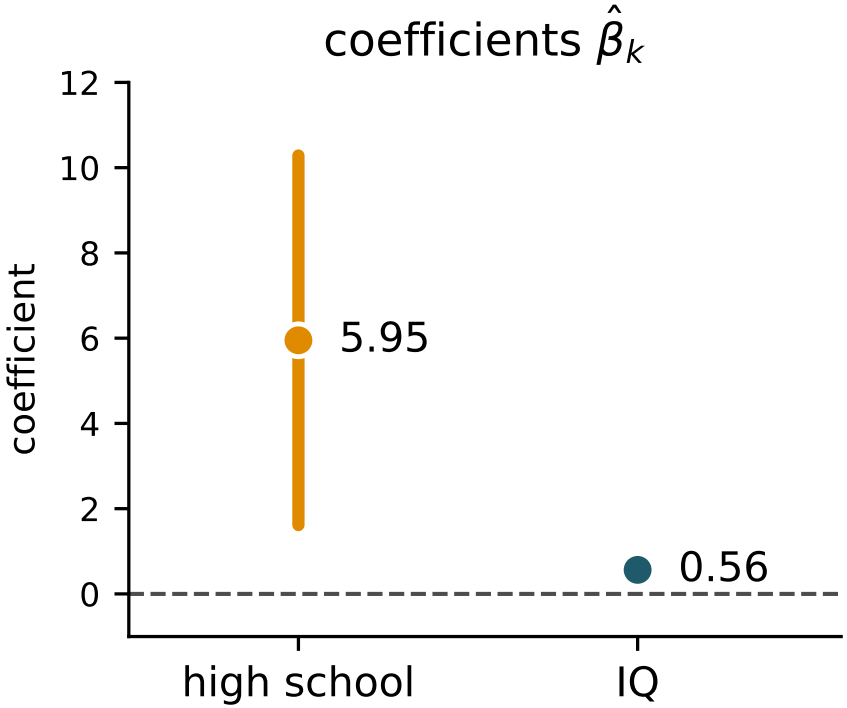
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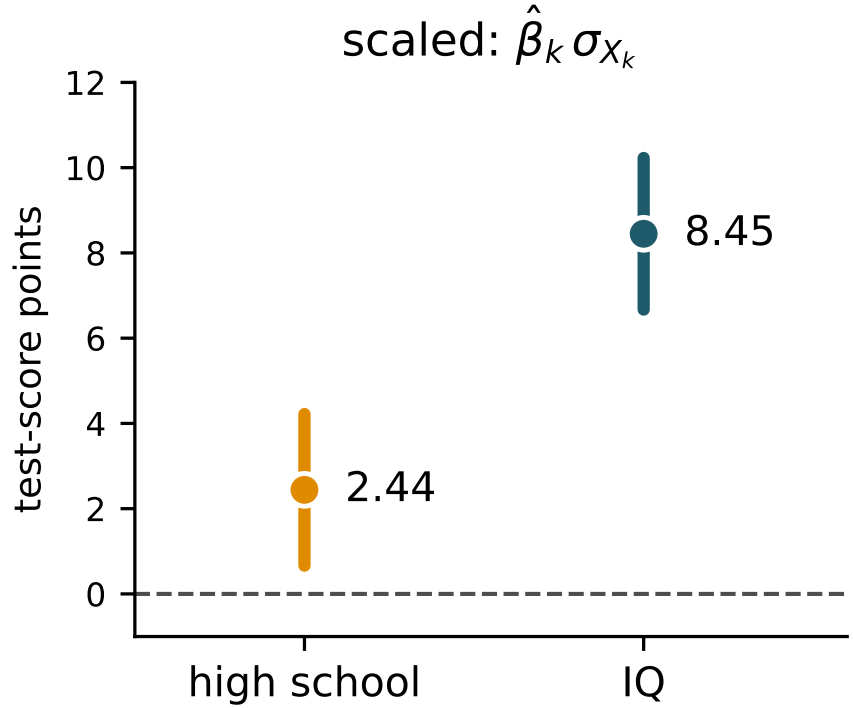
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A one-standard-deviation difference in IQ (15 points) goes with about 8.4 more points on the test; one in schooling ($\sigma_{\text{HS}} = 0.41$) with only about 2.4. IQ matters more.

Raw vs. scaled coefficients

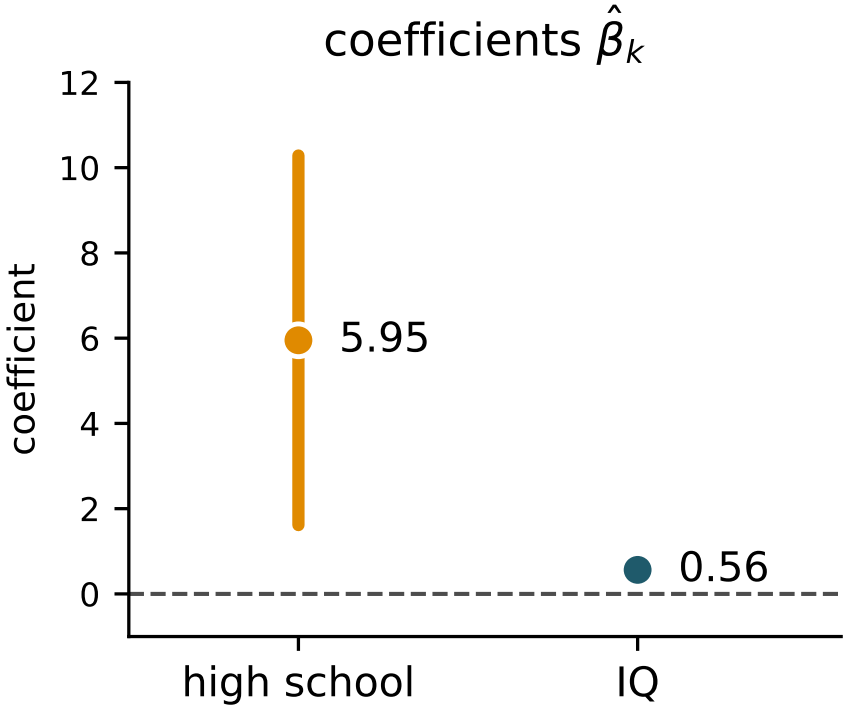
Each coefficient with its 95% confidence interval, before and after multiplying by the standard deviation of its predictor:

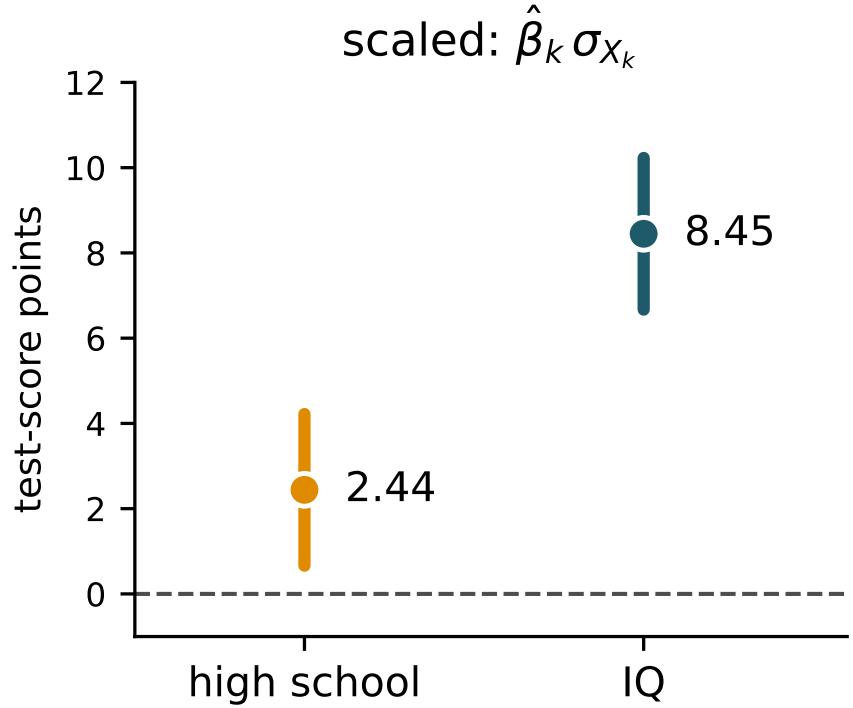




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R^2 with multiple predictors

R^2 is defined exactly as before,

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}}, \qquad \text{SSR} = \sum_{i=1}^{N} (Y_i - \hat{Y}_i)^2, \qquad \text{SST} = \sum_{i=1}^{N} (Y_i - \overline{Y})^2,$$

where now the fitted value $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \cdots + \hat{\beta}_K X_{K,i}$ uses all K predictors ($X_{k,i}$ is the value of predictor k for data point i). It is the fraction of the variation in Y explained by *all* the predictors together:

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`statsmodels` reports it at the top of the summary:

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R^2 for the test scores

$R^2 = 1 - \text{SSR}/\text{SST}$ for three models of the test scores:

predictors	R^2
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- About 79% of the variation is *not* explained: the scores have standard deviation 20, and the residuals still have $\hat{\sigma}_\epsilon \approx 18$. The model says little about any child's score.

Question

Midterm 1 scores for $N = 64$ students in this class: y is the score (fraction of points), `difficult` $= 1$ if you rated the difficulty 4 (the highest rating given) in the week 1 survey, and `QSS` $= 1$ for QSS majors.

	coef	std err	t	P>|t|	[0.025	0.975]
const	0.8705	0.033	26.343	0.000	0.804	0.937
difficult	-0.1002	0.040	-2.507	0.015	-0.180	-0.020
QSS	-0.1340	0.040	-3.361	0.001	-0.214	-0.054

```
print(f"{np.std(difficult):.3f}")
print(f"{np.std(QSS):.3f}")
0.498
0.499
```

	difficult		
	0	1	total
QSS = 0	0.281	0.188	0.469
QSS = 1	0.266	0.266	0.531
total	0.547	0.453	1

Which is the more important predictor of the midterm score? What is the overall regression coefficient (if no other predictors are included) of difficulty?

Solution: midterm scores

More important predictor. Both predictors are 0/1 with standard deviation ≈ 0.5, so scaling barely changes anything:

$$\hat{\beta}_{\text{diff}}\,\sigma_{\text{diff}} = -0.100 \times 0.498 = -0.050, \quad \hat{\beta}_{\text{QSS}}\,\sigma_{\text{QSS}} = -0.134 \times 0.499 = -0.067.$$

QSS is (slightly) more important; here we can compare the raw coefficients directly.

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QSS is (slightly) more important; here we can compare the raw coefficients directly.

Single-predictor coefficient of difficulty. $\beta'_{\text{diff}} = \beta_{\text{diff}} + \beta_{\text{QSS}}\,\beta_{\text{diff,QSS}}$, where from the table

$$\begin{aligned}\hat{\beta}_{\text{diff,QSS}} &= E[X_{\text{QSS}} \mid X_{\text{diff}} = 1] - E[X_{\text{QSS}} \mid X_{\text{diff}} = 0] \\ &= \frac{0.266}{0.453} - \frac{0.266}{0.547} = 0.587 - 0.486 = 0.101,\end{aligned}$$

so $\hat{\beta}'_{\text{diff}} = -0.100 + (-0.134)(0.101) = -0.114$. (Fitting y on `difficult` alone gives -0.1137.)